```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
try:
    df = pd.read_csv("/content/Salary_Data.csv")
except FileNotFoundError:
    from google.colab import files
    uploaded = files.upload()
    df = pd.read_csv(next(iter(uploaded)))
print("First five rows:")
print(df.head())
print("\nDataset information:")
print(df.info())
X = df[["YearsExperience"]]
y = df["Salary"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_predictions = linear_model.predict(X_test)
print("\n--- Linear Regression ---")
print("Slope:", linear_model.coef_[0])
print("Intercept:", linear_model.intercept_)
print("MAE:", mean_absolute_error(y_test, linear_predictions))
print("MSE:", mean_squared_error(y_test, linear_predictions))
print("RMSE:", np.sqrt(mean_squared_error(y_test, linear_predictions)))
print("R2 Score:", r2_score(y_test, linear_predictions))
plt.figure(figsize=(8, 5))
plt.scatter(X, y, color="blue", label="Actual Salary")
plt.plot(
    X, linear_model.predict(X),
    color="red", label="Linear Regression"
)
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Simple Linear Regression")
plt.legend()
plt.grid(True)
plt.show()
poly = PolynomialFeatures(degree=2, include_bias=False)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_predictions = poly_model.predict(X_test_poly)
print("\n--- Polynomial Regression (Degree 2) ---")
print("MAE:", mean_absolute_error(y_test, poly_predictions))
print("MSE:", mean_squared_error(y_test, poly_predictions))
print("RMSE:", np.sqrt(mean_squared_error(y_test, poly_predictions)))
print("R2 Score:", r2_score(y_test, poly_predictions))
X_curve = pd.DataFrame({
    "YearsExperience": np.linspace(
        X["YearsExperience"].min(),
        X["YearsExperience"].max(),
        200
    )
})
y_curve = poly_model.predict(poly.transform(X_curve))
plt.figure(figsize=(8, 5))
plt.scatter(X, y, color="blue", label="Actual Salary")
plt.plot(
    X_curve["YearsExperience"], y_curve,
    color="green", label="Polynomial Regression (Degree 2)"
)
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Polynomial Regression")
plt.legend()
plt.grid(True)
plt.show()
results = pd.DataFrame({
    "Model": ["Linear Regression", "Polynomial Regression"],
    "MAE": [
        mean_absolute_error(y_test, linear_predictions),
        mean_absolute_error(y_test, poly_predictions)
    ],
    "RMSE": [
        np.sqrt(mean_squared_error(y_test, linear_predictions)),
        np.sqrt(mean_squared_error(y_test, poly_predictions))
    ],
    "R2 Score": [
        r2_score(y_test, linear_predictions),
        r2_score(y_test, poly_predictions)
    ]
})

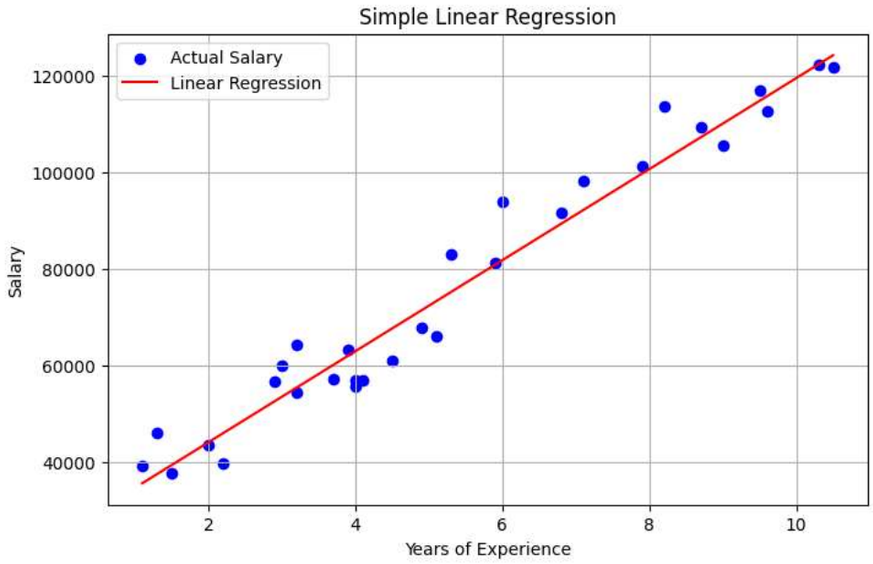
print("\n--- Model Comparison ---")
print(results.to_string(index=False))
years = float(input("\nEnter years of experience: "))
new_data = pd.DataFrame({"YearsExperience": [years]})
linear_salary = linear_model.predict(new_data)[0]
poly_salary = poly_model.predict(poly.transform(new_data))[0]
```

```
print(f"\nLinear Regression predicted salary: {linear_salary:,.2f}")
print(f"Polynomial Regression predicted salary: {poly_salary:,.2f}")
```

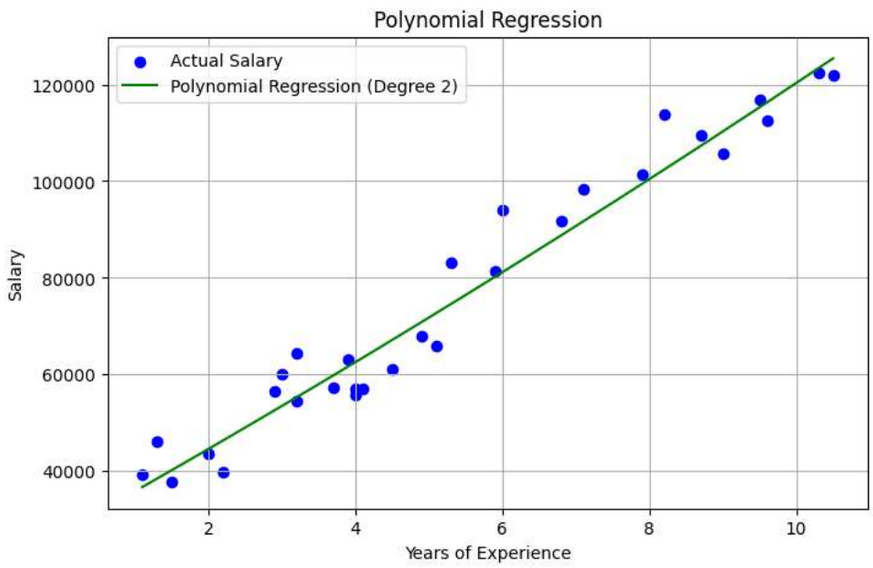
```
First five rows:
   YearsExperience  Salary
0              1.1   39343
1              1.3   46205
2              1.5   37731
3              2.0   43525
4              2.2   39891

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
None

--- Linear Regression ---
Slope: 9423.815323030976
Intercept: 25321.583011776813
MAE: 6286.453830757749
MSE: 49830096.85590839
RMSE: 7059.04362190151
R2 Score: 0.9024461774180497
```




```
--- Polynomial Regression (Degree 2) ---
MAE: 6381.371429702194
MSE: 52527916.36877497
RMSE: 7247.61452953832
R2 Score: 0.897164578089087
```




```
--- Model Comparison ---
                Model         MAE         RMSE  R2 Score
    Linear Regression 6286.453831 7059.043622  0.902446
Polynomial Regression 6381.371430 7247.614530  0.897165

Enter years of experience: 5

Linear Regression predicted salary: 72,440.66
Polynomial Regression predicted salary: 71,791.02
```